

Learned memory control for long conversations: what we set out to do, and what we found

One-page summary · 7 October 2026 · Details and every number: EXPERIMENTS.md §1–§19d and docs/research/THESIS_RESULTS_OUTLINE.md

AIM

A language model answering questions about a long history (weeks of chat) cannot read all of it at once. A **memory controller** decides what stays in view, what goes to an archive, and what is searched back when a question arrives. We asked whether a **learned** controller (imitation of a hindsight teacher, then reinforcement learning) does better than simple rules at the same token budget. We tested on synthetic tasks, then on two real benchmarks (LoCoMo, LongMemEval), with a frozen 7B reader (Qwen2.5-7B) and an official judge. Every paid test was pre-registered: its success rule was written down and passed a free check before it ran.

RESEARCH QUESTIONS

- **RQ1.** Does a learned controller beat simple rules under the same budget?
- **RQ2.** Can a controller learn from hindsight (what later turned out to be needed)?
- **RQ3.** Does reinforcement learning (GRPO) add anything to imitation?
- **RQ4.** Does it hold up with real models and conversations, and when is it worth it?

WHAT WE DID, AND WHY IT WORKED OR NOT

Step	Result	Why
Synthetic tasks: learn from a hindsight “regret” teacher plus a follow-the-clue search (§10–11)	Worked	Beats every rule, closes about half the gap to the hindsight oracle. Needs are structured and predictable.
Real chats: learn what to keep <i>before</i> the question (§13)	Did not	Kept more evidence (+0.09–0.13) but no accuracy gain: the reader worsens past ~3k prompt tokens (“context rot”).
Charge a token price for kept memory (§14)	No-go	Before the question, a turn’s later need is barely predictable (AUC 0.62–0.74).
Learn <i>at the question</i> which 5 of 16 search hits to show (§15)	Worked	0.513 vs 0.466 for the best rule, at half the tokens (745 vs 1,554); replicated over seeds and hosts.
GRPO on that selector, rewarded by the reader (§16)	Did not	+0.013 (CI –0.005, +0.033). 44–61% of sample groups tie; each sample carries too little signal.
Second-family judge, Granite-8B (§18)	Confirms	97% agreement (κ 0.95), similar error rates; same ranking.
Show 8 instead of 5 turns (§19c)	Worked	0.545 vs 0.500 (+0.045, CI +0.011, +0.079) at 1,049 tokens; beats FIFO by +0.113 at a third of its tokens.
Choose how many turns per question (§19, §19b)	Did not	A perfect chooser would halve the tokens at equal evidence; our two choosers could not tell easy questions from hard ones well enough.

ANSWERS

- **RQ1: Yes, but only at the question.** On synthetic tasks a learned controller wins clearly. On real conversations, deciding what to keep beforehand does not beat rules; choosing which retrieved turns the reader sees does: +0.047 over the best rule, then +0.045 more by showing eight turns.
- **RQ2: On synthetic tasks yes; on real chats, not enough.** Hindsight says what was needed, but on real text that cannot be predicted before the question is asked. The useful signal arrives with the question.
- **RQ3: At most about one point.** The limit is how much each reader-graded sample tells the learner, not how credit is split over time.
- **RQ4: Yes, with a real 7B reader and two judges;** it is worth it when the prompt stays small. With the same selector, 16 turns put more evidence in view than 8 but scored –0.057 (CI –0.096, –0.019): with the evidence present, accuracy fell from 0.68 to 0.58 at 2.6k tokens. A pre-registered repeat (second head, second host) gave –0.026 (CI –0.066, +0.015), not significant, but the same fall with the evidence present (0.71 to 0.62). LongMemEval is a *selection* problem; LoCoMo is a *search* problem (evidence spread over many short turns).

LIMITATIONS

- One reader and one frozen prompt; the ~3k-token onset of context rot is measured for that pair only.
- The positive reader result is pre-registered on LongMemEval only. The LoCoMo pick (5 turns) missed (+0.023, CI –0.001, +0.048). The LoCoMo win of 16 turns over FIFO (0.344 vs 0.295 at a quarter of the tokens) is exploratory.
- The context-rot accuracy loss was significant in one of two runs; its confirmation missed. Both judges are 8B-class.
- LoCoMo intervals rest on only 10 conversations. The §15 arm was the best of three before it was replicated.

FUTURE WORK

- A larger, pre-registered test of the context-rot pair (more questions or seeds), since one confirmation missed.
- A second reader of another family or size, to see whether the onset moves.
- A chooser that reaches the per-question ceiling (half the tokens at equal evidence).
- LoCoMo as a search problem: better retrieval units (fact keys, session-level search) and heads trained on LoCoMo.
- A leave-one-out set reward for GRPO (declared, not run).

Total GPU spend for all experiments: about \$8.63 (RunPod A40, every launch approved in advance). Accuracy is the share of questions the judge marks correct; intervals are paired 95% bootstrap intervals.